Fall 1993

Problem 1 (Fa93). Let X be a metric space and (x_n) a convergent sequence in X with limit x_0. Prove that the set $C = \{x_0, x_1, x_2, \ldots\}$ is compact.

Problem 2 (Fa87, Fa93). Let A be the group of rational numbers under addition, and let M be the group of positive rational numbers under multiplication. Determine all homomorphisms $\varphi : A \to M$.

Problem 3 (Fa93). Describe the region in the complex plane where the infinite series

$$\sum_{n=1}^{\infty} \frac{1}{n^2} \exp\left(\frac{nz}{z-2}\right)$$

converges. Draw a sketch of the region.

Problem 4 (Fa93). Let $\mathbb{F}$ be a field. Fix m and n, and fix matrices A and B in $M_{m\times n}$. Define the linear transformation $T : M_{n\times m} \to M_{m\times n}$ by

$$T(X) = AXB$$

Prove that if $m \neq n$, then T is not invertible.

Problem 5 (Fa93). Let the function $x(t)$ for $-\infty < t < \infty$ be a solution of the differential equation

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} - 2b\,\frac{\mathrm{d}x}{\mathrm{d}t} + cx = 0$$

such that $x(0) = x(1) = 0$ and b and c are real constants. Prove that $x(n) = 0$ for every integer n.

Problem 6 (Su79, Fa93). Let A, B, and C be finite abelian groups such that $A \times B$ and $A \times C$ are isomorphic. Prove that B and C are isomorphic.

Problem 7 (Sp79, Fa93). Let $M_{n\times n}$ denote the vector space of $n \times n$ real matrices for $n \geqslant 2$. Let $\det : M_{n\times n} \to \mathbb{R}$ be the determinant map.

1. Show that det is C^∞.

2. Show that the derivative of det at $A \in M_{n\times n}$ is zero if and only if A has rank $\leqslant n-2$.

Problem 8 (Sp90, Fa93, Sp13, Sp18, Sp19, Sp22).

1. If A is a matrix over an algebraically closed field of characteristic 0 with $A^n = I$ for some $n > 0$ show that A is diagonalizable.

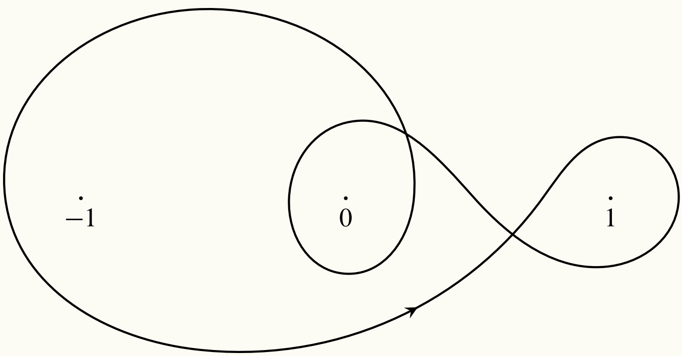
2. Show that over any field of characteristic $p > 0$ there is a matrix A, with $A^n = I$ for some $n > 0$, that is not diagonalizable.

Problem 9 (Fa93). Evaluate $\displaystyle\int_{-\infty}^{\infty} \frac{e^{-ix}}{x^2 - 2x + 4}\,\mathrm{d}x$.

Problem 10 (Fa93). Let f be a continuous real valued function on $[0, \infty)$. Let A be the set of real numbers a that can be expressed as $a = \lim_{n\to\infty} f(x_n)$ for some sequence (x_n) in $[0, \infty)$ such that $\lim_{n\to\infty} x_n = \infty$. Prove that if A contains the two numbers a and b, then contains the entire interval with endpoints a and b.

Problem 11 (Fa93). Let R be a commutative ring with identity. Let G be a finite subgroup of R^*, the group of units of R. Prove that if R is an integral domain, then G is cyclic.

Problem 12 (Fa93). Evaluate the integral $\displaystyle\frac{1}{2\pi i}\int_\gamma \frac{e^z}{z^2(1-z^2)}\,\mathrm{d}z$ one the curve γ depicted by



Problem 13 (Sp90, Fa93, Sp94). Show that there are at least two nonisomorphic non-abelian groups of order 24, of order 30 and order 40.

Problem 14 (Fa93, Fa77). Let n be an integer larger than 1. Is there a differentiable function on $[0, \infty)$ whose derivative equals its n-th power and whose value at the origin is positive?

Problem 15 (Su79, Fa93). Prove that the matrix

$$\begin{pmatrix} 0 & 5 & 1 & 0 \\ 5 & 0 & 5 & 0 \\ 1 & 5 & 0 & 5 \\ 0 & 0 & 5 & 0 \end{pmatrix}$$

has two positive and two negative eigenvalues (counting multiplicities).

Problem 16 (Fa93). Let K be a continuous real valued function defined on $[0, 1] \times [0, 1]$. Let $\mathcal{F}$ be the family of functions f on $[0, 1]$ of the form

$$f(x) = \int_0^1 g(y)K(x, y)\,\mathrm{d}y$$

with g a real valued continuous function on $[0, 1]$ satisfying $|g| \leqslant 1$ everywhere. Prove that the family $\mathcal{F}$ is equicontinuous.

Problem 17 (Fa93, Sp24). Let w be a positive continuous function on $[0, 1]$, n a positive integer, and P_n the vector space of real polynomials whose degrees are at most n, equipped with the inner product

$$\langle p, q \rangle = \int_0^1 p(t)q(t)w(t)\,\mathrm{d}t$$

1. Prove that P_n has an orthonormal basis $p_0, p_1, \ldots, p_n$, i.e., $\langle p_j, p_k \rangle = 1$ for $j = k$ and 0 for $j \neq k$ such that deg $p_k = k$ for each k.

2. Prove that $\langle p_k, p_k' \rangle = 0$ for each k.

Problem 18 (Fa93). Let f be a continuous real valued function on $[0, 1]$, and let the function h in the complex plane be defined by

$$h(z) = \int_0^1 f(t) \cos zt\,\mathrm{d}t$$

1. Prove that h is analytic in the entire plane.

2. Prove that h is the zero function only if f is the zero function.